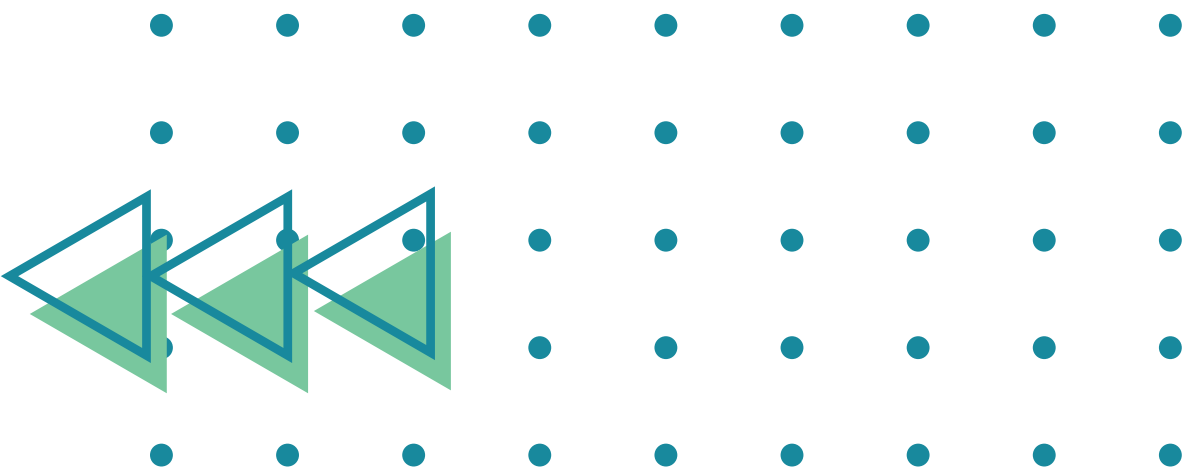
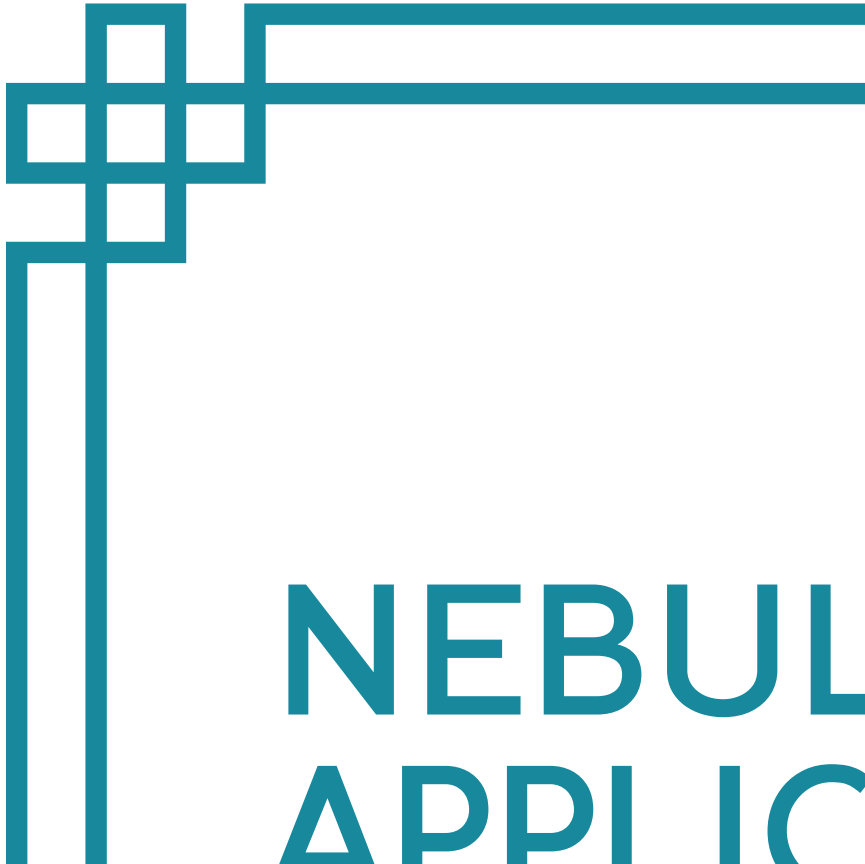
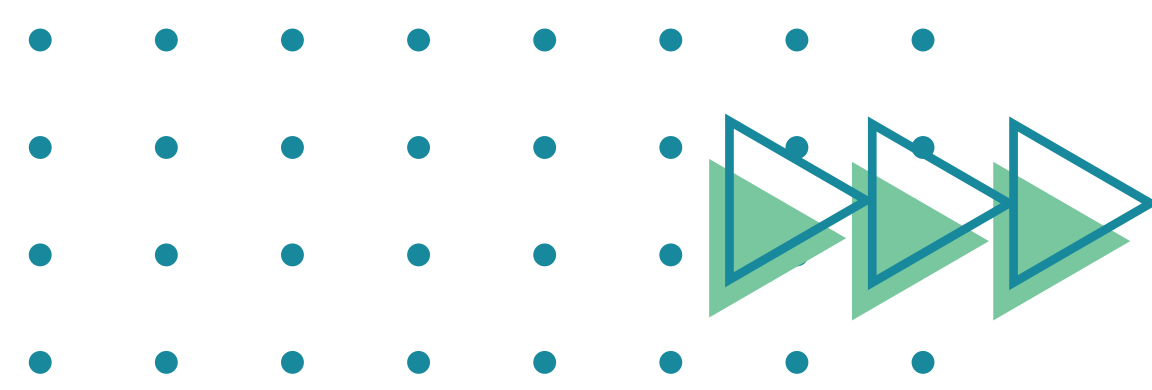
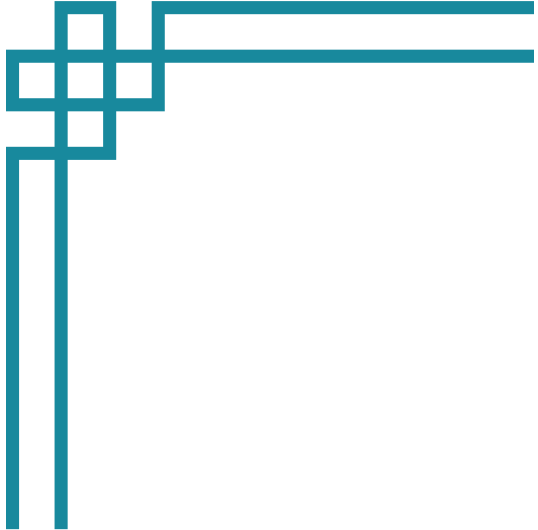




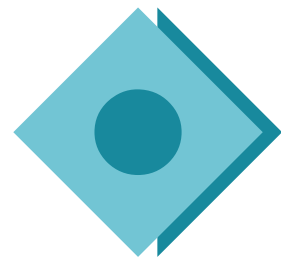
NEBULA LEARN - TOPRANK WEB APPLICATION DOCUMENTATION AND SYSTEM OVERVIEW

By ADYTIA AGUSTIAWAN | Bachelor of Information
Technology, Semester 2 | Universitas Amikom Yogyakarta

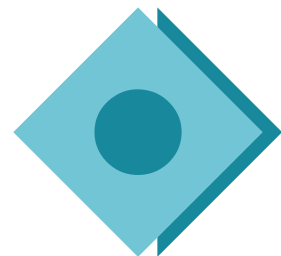




PROJECT & PARTICIPANT IDENTITY



Name: ADYTIA AGUSTIAWAN
Major: Bachelor of Information Technology
Semester: 2
NIM: 25.60.0209
University: Universitas Amikom Yogyakarta



Project: TopRank Web Application
Deployment: Vercel Cloud Platform
Project URL: <https://nebulalearn-umber.vercel.app/>
Status: Active & Deployed



APPLICATION OVERVIEW

Dual-Role System: Admin & Student (Mahasiswa) — Core functionalities include skill and portfolio verification, leaderboard and reward management, responsive design with Docker deployment, and AI-powered recommendations.

Key Features: Submit and verify skills/portfolios, track leaderboard rankings by points, browse and claim rewards, containerized deployment, and intelligent AI-driven suggestions for skills and achievements.

Purpose: Facilitate student skill validation and reward achievements — enabling transparent, data-driven recognition of academic and technical competencies at Universitas Amikom Yogyakarta.

SYSTEM ARCHITECTURE & ROLES



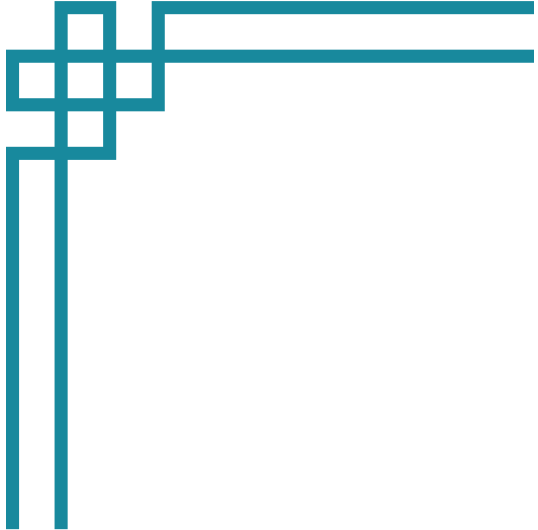
Admin Module

- Authentication: Login & Logout
- Dashboard with key metrics and visual data
- Student data search by skill and points
- Skill & portfolio verification (approve/reject)
- Reward management and points threshold



Student Module

- Authentication: Login & Logout
- Talent profile: submit skills with evidence
- Portfolio upload and verification tracking
- Leaderboard: rankings by points
- Reward catalog browsing and claiming
- Technical: Responsive UI, Docker, AI features



ADMIN MODULE OVERVIEW



Authentication & Dashboard: Secure admin login and logout with session handling and protected routes. Dashboard displays key metrics including total student count, registered skills, verified portfolios, and visual data summaries for quick monitoring.

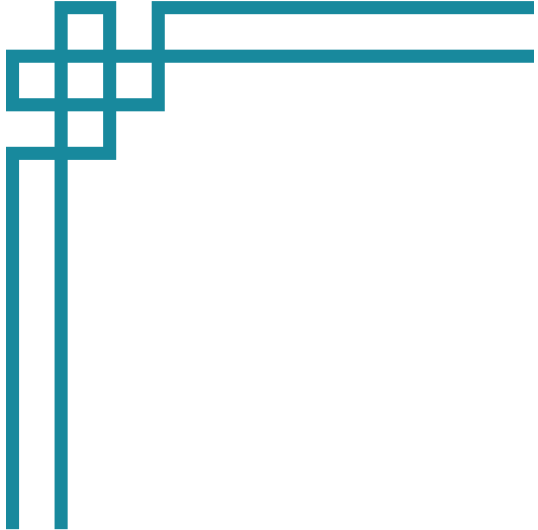


Student Data & Verification: Search and filter student data by skill or points using a quick-access search bar. Review, approve, or reject pending skill submissions and portfolio verifications with full status tracking (pending / approved / rejected).

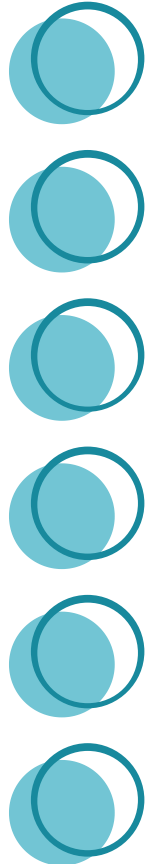
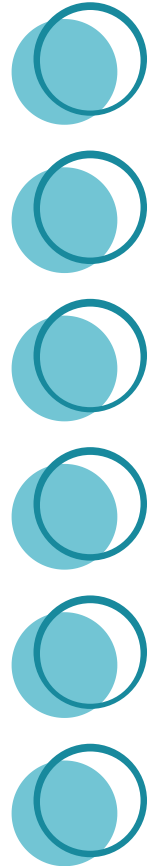


Reward Management: Create and publish new rewards with configurable minimum points thresholds. Manage the full reward catalog and inventory, ensuring students can browse and claim rewards based on their accumulated achievement points.





ADMIN AUTHENTICATION



Secure Admin Login & Session Handling

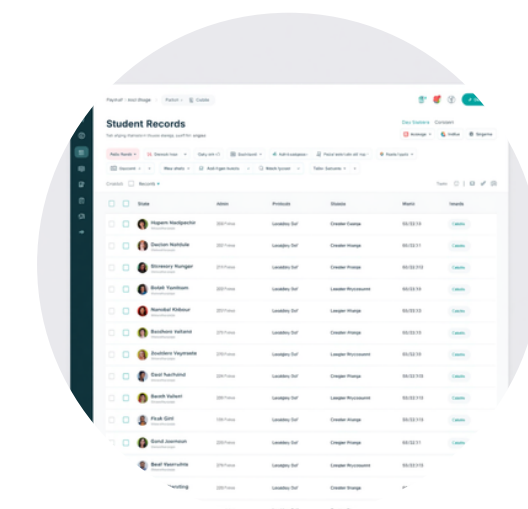
- Protected login page with credential validation
- Session-based authentication on successful login
- Admin-only route protection (unauthorized redirect)
- Persistent session management across pages
- Screenshot: Admin login form and dashboard access

Logout & Session Termination

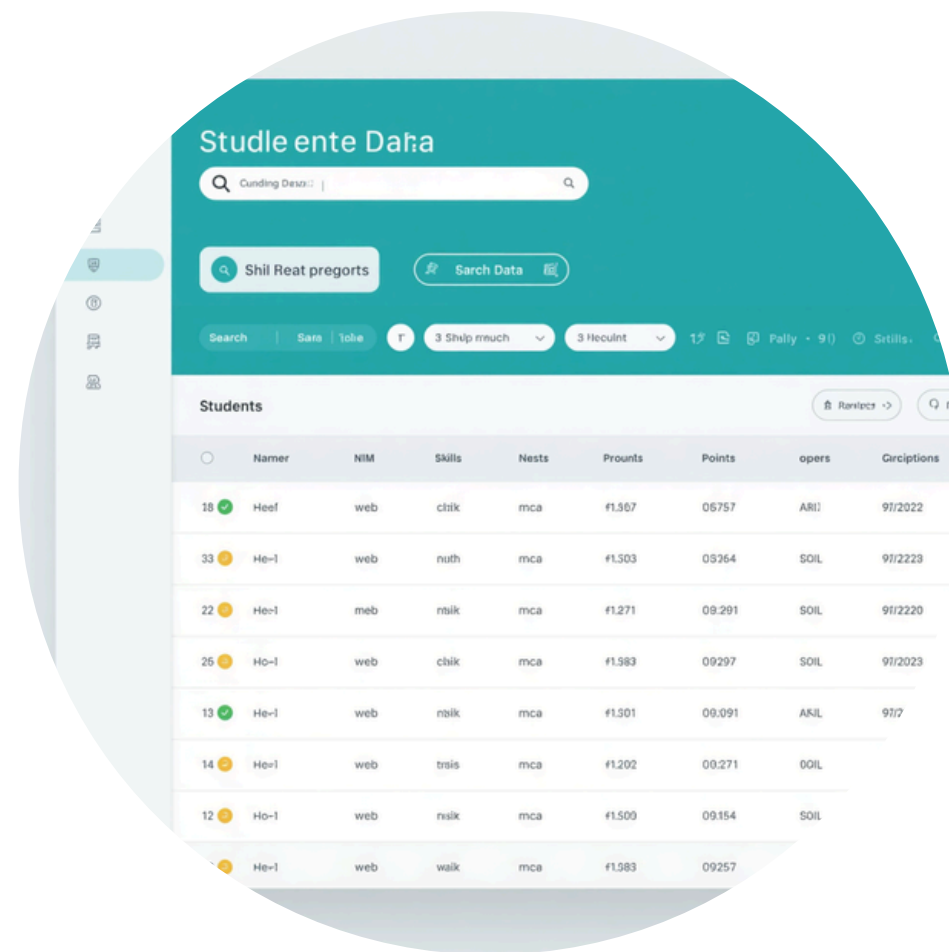
- Logout functionality clears active session data
 - Redirects to login page upon session end
 - Prevents re-access to protected routes post-logout
 - Secure session invalidation on server side
 - Screenshot: Logout button and session termination flow
- 

A decorative graphic element in the top-left corner of the page. It consists of a 3x3 grid of squares. The top-left square is white, while the other eight squares are filled with a teal color. From the top-right corner of the grid, a horizontal line extends to the right edge of the page. From the bottom-left corner of the grid, a vertical line extends to the bottom edge of the page. The lines are teal and have a consistent thickness.

-



ADMIN - STUDENT DATA SEARCH

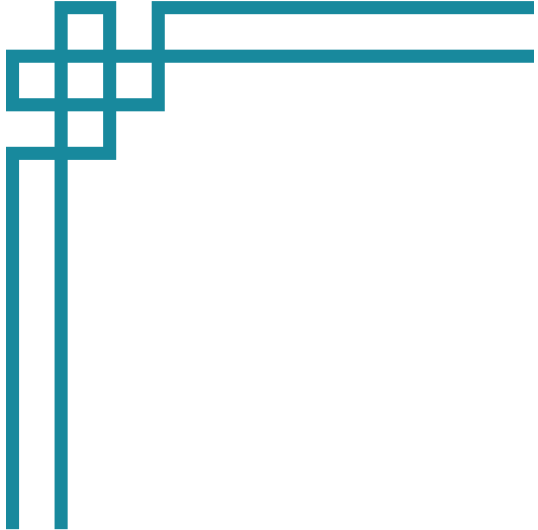


The screenshot displays a web application interface for searching student data. At the top, there's a header 'Stuđente Data' with a search bar containing 'Cunding Deso...'. Below the header, there are buttons for 'Shil Reat pregorts' and 'Sarch Data'. A navigation bar includes 'Search', 'Save', 'Take', and several dropdown menus. The main content area is titled 'Students' and contains a table with the following columns: Name, NIM, Skills, Nests, Prounts, Points, opers, and Circipions. The table lists several student records with their respective details.

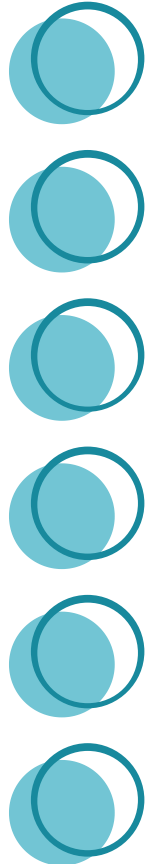
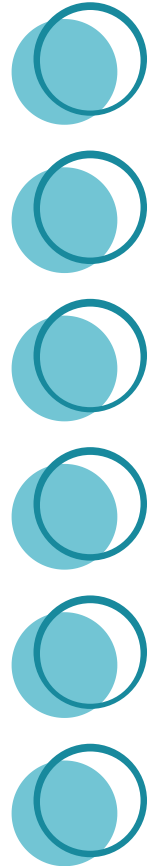
	Name	NIM	Skills	Nests	Prounts	Points	opers	Circipions
18	Heef	web	chik	mca	#1.567	05757	ARIJ	9/1/2022
33	He-l	web	nuth	mca	#1.503	03764	SOIL	9/1/2223
22	He-l	web	mlk	mca	#1.271	09.291	SOIL	9/1/2220
26	Ho-l	web	chik	mca	#1.583	09297	SOIL	9/1/2023
13	He-l	web	mlk	mca	#1.501	09.091	AKIL	9/1/7
14	He-l	web	trls	mca	#1.202	09.271	OGIL	
12	Ho-l	web	mlk	mca	#1.509	09.154	SOIL	
	He-l	web	mlk	mca	#1.583	09257		

Student data table/list view with comprehensive search and filter capabilities. Quickly locate students by filtering based on registered skill categories or accumulated points threshold. The search interface provides real-time results for efficient student management.

Quick search bar enables rapid access to any student record by name or NIM. Filter combinations allow admins to cross-reference skill and points data simultaneously, supporting streamlined verification workflows and reward eligibility checks.



ADMIN - SKILL & PORTFOLIO VERIFICATION



Skill Submission Review

- View all pending skill submissions from students
- Approve verified skills with evidence confirmation
- Reject invalid submissions with feedback notes
- Status tracking: Pending → Approved / Rejected
- Each skill entry includes proof links or uploaded files

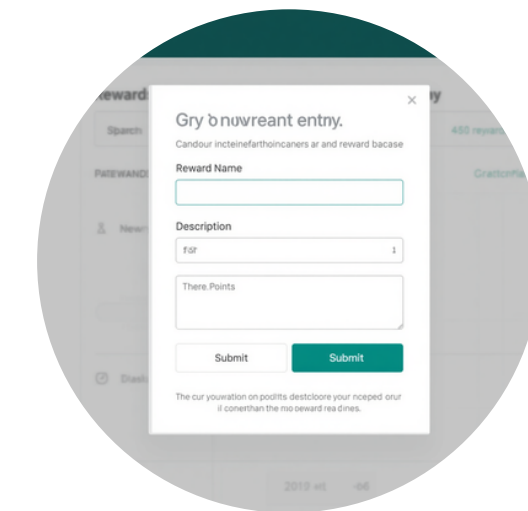
Portfolio Verification

- Review student-submitted portfolio projects
 - Verify attached proof files, links, or images
 - Approve or reject portfolio entries accordingly
 - Real-time status update visible to students
 - Full audit trail: before and after approval states
- 

ADMIN - REWARD MANAGEMENT

Create and post new rewards for students who meet achievement thresholds. Set minimum points required for each reward, define eligibility criteria, and publish rewards to the catalog. Admins can edit or remove existing rewards at any time to keep the catalog current and relevant.

Manage the full reward catalog and inventory from the admin panel. Monitor claimed rewards, track remaining stock, and review claim requests. The system ensures only eligible students (meeting point thresholds) can successfully claim available rewards.

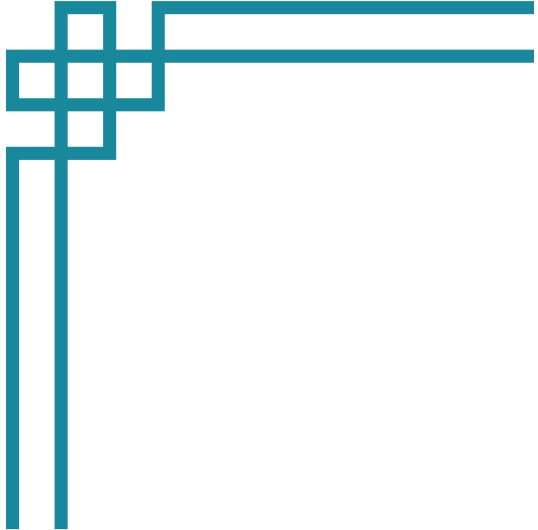


STUDENT MODULE OVERVIEW

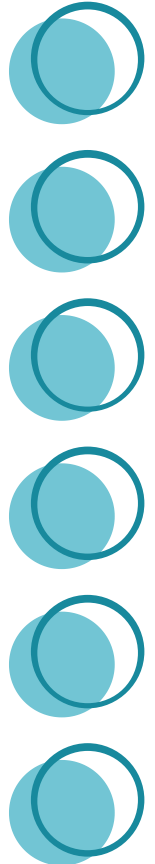
Authentication (Login & Logout): Secure student login with protected routes and session management. Talent Profile: Create and update skills, submit for verification with evidence (links/files/images), and track submission status (pending/approved/rejected).

Portfolio Management: Upload and submit portfolios for admin verification. Attach proof or evidence files and links. Track verification status in real time through the student dashboard.

Leaderboard & Rewards: View student rankings by points on the leaderboard. Browse the reward catalog, check points requirements, claim eligible rewards, and track claim status (claimed/pending).



STUDENT AUTHENTICATION AND TALENT PROFILE



Student Login & Logout

- Secure student login page with session handling
- Protected routes accessible only after authentication
- Logout functionality with full session termination
- Redirect to login on unauthorized access attempts

Talent Profile Management

- Create and update personal skill profile
 - Submit skills with proof (links, files, or images)
 - Track submission status: Pending / Approved / Rejected
 - Portfolio upload linked to verification workflow
- 

[illegible]

LEADERBOARD & REWARD CLAIMING



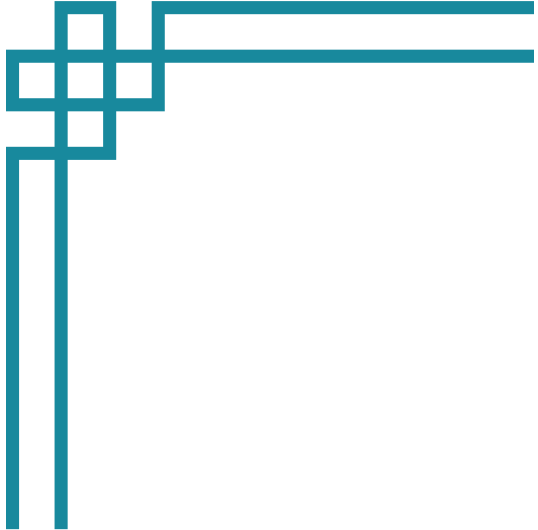
Leaderboard

- Students ranked by total points earned
- Displays name, rank, and point score
- Real-time ranking updates on skill/portfolio approval
- Top performers highlighted at the top
- Motivates students to earn more verified skills
- Screenshot: Leaderboard page with ranked student list



Reward Catalog & Claiming

- Browse available rewards with point requirements
- Claim rewards when eligible (points threshold met)
- Claim status tracking: Pending / Claimed
- Admin-managed reward inventory
- Students notified upon successful claim
- Screenshot: Reward catalog with claim button visible



TECHNICAL MODULE OVERVIEW



Responsive Layout: Fully responsive UI supporting mobile, tablet, and desktop screen sizes. Adaptive design ensures optimal user experience across all devices and resolutions.



Dockerized Deployment: Containerized local deployment using Docker. Includes build and run commands, port mapping configuration, and isolated environment setup for consistent performance.



Vercel & AI Recommendations: Live deployment on Vercel at <https://nebulalearn-umber.vercel.app/>. AI-powered recommendation engine suggests skills, portfolio improvements, and rewards based on user profile and points.



RESPONSIVE DESIGN & DOCKER DEPLOYMENT

Responsive Layout Support

The TopRank web application is fully responsive across mobile, tablet, and desktop devices. Flexible grid layouts and media queries ensure consistent UX on all screen sizes. UI components adapt dynamically to viewport width, providing seamless navigation and readability whether accessed from a smartphone or a widescreen monitor.



Docker Deployment

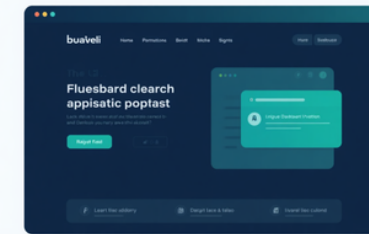
The application is containerized using Docker for consistent local deployment. Build command: `docker build -t nebulalearn .` | Run command: `docker run -p 3000:3000 nebulalearn` | Port mapping: `localhost:3000 → container:3000`. Docker ensures environment consistency across development and production, eliminating dependency conflicts.



VERCEL DEPLOYMENT & AI RECOMMENDATIONS

Live deployment hosted on Vercel at <https://nebulalearn-umber.vercel.app/>. The application is publicly accessible with zero-downtime deployment, automatic HTTPS, and CI/CD integration. Any push to the main branch triggers an automatic redeploy, ensuring the latest version is always live.

AI-powered recommendation engine analyzes each student's profile, submitted skills, portfolio content, and accumulated points to deliver personalized suggestions. Recommendations include skill development paths, portfolio improvement tips, and eligible rewards — helping students maximize their learning outcomes.

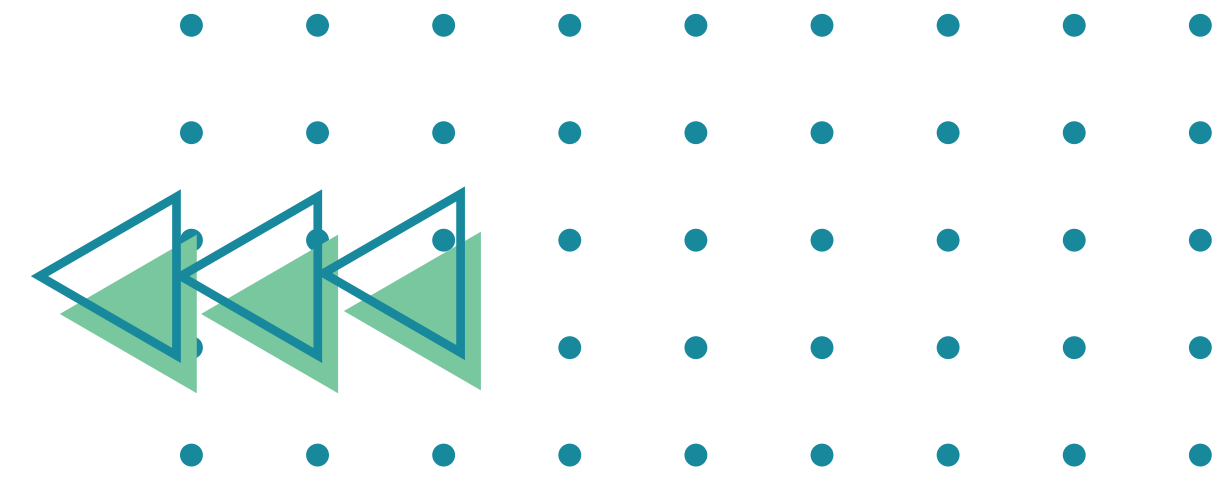
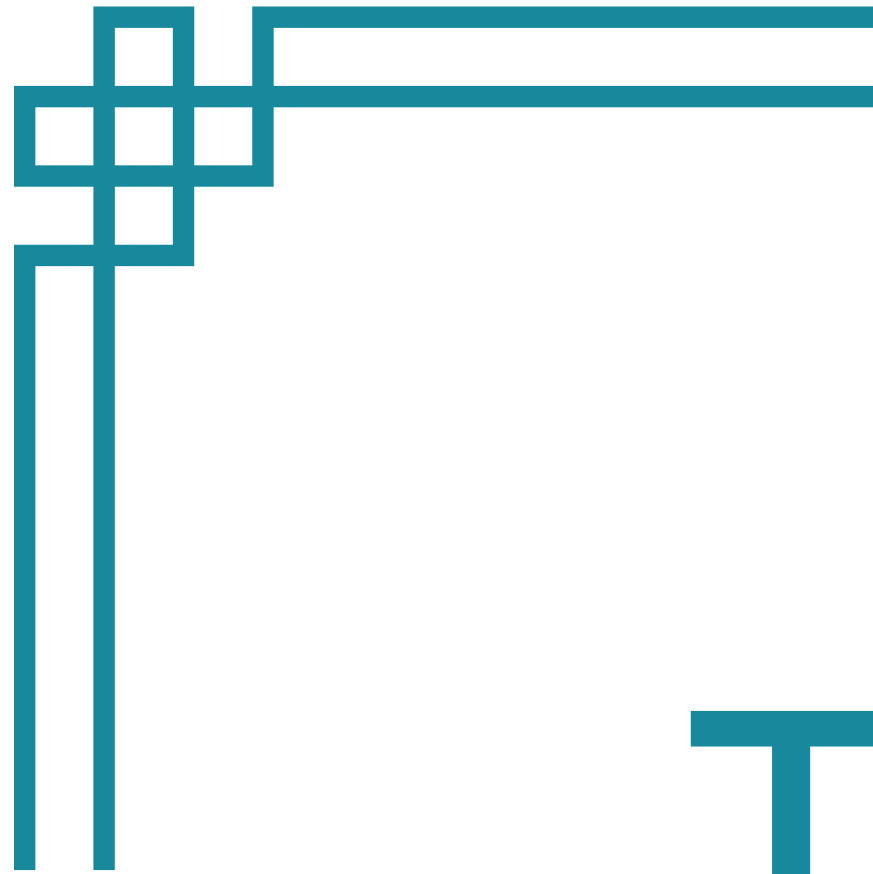


CRITERIA CHECKLIST & COMPLIANCE

✓ Admin: Login/Logout ✓ Dashboard with Metrics ✓ Student Data Search ✓ Skill & Portfolio Verification ✓ Reward Management

✓ Student: Login/Logout ✓ Talent Profile & Skill Submission ✓ Portfolio Upload & Tracking ✓ Leaderboard Rankings ✓ Reward Catalog & Claiming

✓ Responsive Design (Mobile/Tablet/Desktop) ✓ Docker Containerization & Deployment ✓ Vercel Live Deployment ✓ AI-Powered Recommendations — All 14 criteria implemented with screenshot proof.



THANK YOU

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